

Prescribed-Time Adaptive Position and Attitude Tracking Control for Patrol Quadrotor UAV

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Abstract: This paper addresses the challenging problem of high-precision and robust trajectory tracking control for a quadrotor unmanned aerial vehicle (QUAV) operating in complex power grid environments. First, an Euler-angle-based QUAV model considering external disturbances is established. Then, a novel non-piecewise time-varying gain function is constructed by introducing polynomial and exponential terms, thereby mitigating the control abruptness and non-smoothness inherent in traditional gain functions. Based on this, an adaptive prescribed-time controller is systematically proposed to guarantee the rapid convergence of tracking errors irrespective of initial conditions, which enhances inspection efficiency. Moreover, an adaptive update law is designed to online estimate and compensate for lumped disturbances, including inherent model uncertainties and external environmental perturbations. Finally, the closed-loop stability is rigorously established via Lyapunov theory, and the effectiveness of the proposed control strategy is verified through numerical simulations.

Key Words: Quadrotor UAV, prescribed-time control, adaptive control, position and attitude tracking

1 Introduction

The modernization of power systems necessitates rigorous maintenance of critical assets like substations and high-voltage cables to guarantee grid reliability. Given the increasing scale of transmission networks, traditional manual inspections have proven inadequate [1]. Consequently, QUAVs have emerged as a dominant solution for autonomous inspections, owing to their agility and vertical takeoff and landing characteristics [2, 3]. While this non-contact approach improves efficiency and reduces personnel risk, practical applications face significant hurdles. Specifically, complex external disturbances and inherent model uncertainties impose stringent requirements on control system robustness. Therefore, designing a high-quality control strategy to cope with environmental disturbances and parametric uncertainties remains a key research challenge [4].

Existing research on QUAV control encompasses linear approaches, such as PID [5] and linear quadratic regulators [6], as well as nonlinear techniques like sliding mode control [7, 8] and robust adaptive control [9]. While these methods are effective, they generally guarantee only asymptotic convergence, implying that tracking errors theoretically approach zero only when time tends to infinity. To accommodate the time-critical requirements of power system inspection tasks, finite-time control theories have been introduced [10]. Beyond ensuring stability within a finite duration, the method offers faster convergence rates and enhanced disturbance rejection capabilities compared to asymptotic approaches. Nevertheless, a critical limitation is that the settling time in finite-time control remains heavily dependent

on the initial system states. To mitigate this dependency, fixed-time control algorithms were developed, offering a convergence time upper bound that is insensitive to initial conditions [11–13]. However, this upper bound is often a conservative estimate derived from Lyapunov inequalities and involves complex parameter tuning, which precludes the direct pre-assignment of an exact convergence time. Consequently, prescribed-time control has recently emerged as a superior solution [14, 15]. It allows the convergence time to be explicitly pre-set as a control parameter, independent of initial conditions [16], thus ensuring precise temporal performance.

While combining prescribed-time theory with nonlinear control ensures precise convergence within a pre-assigned time [17], the choice of control structure is critical. The backstepping method [18] stands out as a superior candidate due to its recursive structure and chattering-free performance. However, the existing literature suffers from two main limitations: On the one hand, most existing studies concentrate on the attitude subsystem [19], overlooking the position subsystem that is essential for precise trajectory tracking and hovering. On the other hand, prescribed-time control commonly employs time-varying gain functions that become singular at the terminal time [20], which may trigger impulsive control inputs and actuator saturation, thus limiting practical applicability.

To overcome the aforementioned limitations, this paper constructs a novel non-piecewise time-varying gain function by introducing polynomial and exponential terms. This design effectively mitigates the issues of control singularity and signal discontinuity. Based on this, a prescribed-time control strategy is proposed to ensure tracking errors converge strictly within a user-defined duration, thereby significantly enhancing the inspection efficiency and adaptability of the QUAV. Specifically, an adaptive update law is employed to estimate the lumped disturbances, comprising both endogenous model uncertainties and exogenous environmental disturbances. Finally, the closed-loop stability is

This work was supported in part by the National Natural Science Foundation of China under Grant 62203384, in part by the Zhejiang Provincial Natural Science Foundation of China under Grant LMS25F030014, in part by the Open Research Project of the State Key Laboratory of Industrial Control Technology under Grant ICT2026B54, in part by the Fundamental Research Funds for the Zhejiang University of Science and Technology under Grant 2025QN012, and in part by the Graduate Research and Innovation Fund of Zhejiang University of Science and Technology under Grant 2024yjskc04.

rigorously analyzed using Lyapunov theory, ensuring that the position and attitude tracking errors satisfy practically prescribed-time stability (PPTS).

2 Problem Formulation and Preliminaries

2.1 System Modeling

The QUAUV is actuated by four motor-driven propellers, where maneuvering is achieved by modulating the rotational speeds to generate differential aerodynamic forces and torques. A simplified structure of a QUAUV is illustrated in Fig. 1. To characterize the system's motion, the inertial coordinate system is established to describe the spatial position $\{x, y, z\}$ and the attitude angles $\{\phi, \theta, \psi\}$, where ϕ , θ , and ψ correspond to the roll (rotation about the x-axis), pitch (rotation about the y-axis), and yaw (rotation about the z-axis) angles, respectively. Considering a standard class of a QUAUV dynamics, the simplified mathematical model can be formulated as follows[21].

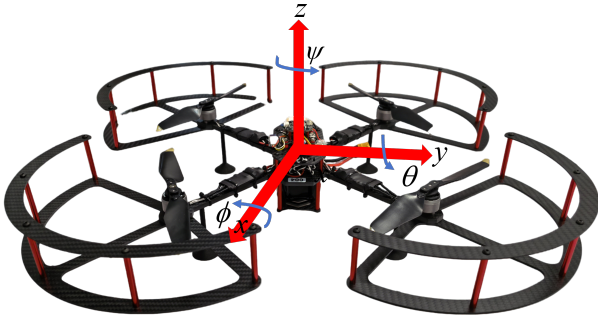


Fig. 1: A simplified structure of a QUAUV.

$$\begin{cases} \ddot{x} = -\frac{k_x}{m}\dot{x} + \frac{\cos\phi\sin\theta\cos\psi + \sin\phi\sin\psi}{m}u_1 + d_x \\ \ddot{y} = -\frac{k_y}{m}\dot{y} + \frac{\cos\phi\sin\theta\sin\psi - \sin\phi\cos\psi}{m}u_1 + d_y \\ \ddot{z} = -\frac{k_z}{m}\dot{z} - g + \frac{\cos\phi\cos\theta}{m}u_1 + d_z \\ \ddot{\phi} = \frac{I_y - I_z}{I_x}\dot{\theta}\dot{\psi} - \frac{k_\phi}{I_x}\dot{\phi} + \frac{1}{I_x}u_\phi + d_\phi \\ \ddot{\theta} = \frac{I_z - I_x}{I_y}\dot{\phi}\dot{\psi} - \frac{k_\theta}{I_y}\dot{\theta} + \frac{1}{I_y}u_\theta + d_\theta \\ \ddot{\psi} = \frac{I_x - I_y}{I_z}\dot{\phi}\dot{\theta} - \frac{k_\psi}{I_z}\dot{\psi} + \frac{1}{I_z}u_\psi + d_\psi \end{cases} \quad (1)$$

where m and g are the total mass and gravitational acceleration of the QUAUV, respectively. I_x , I_y and I_z are the rotary inertia. For $i = x, y, z, \phi, \theta, \psi$, k_i denotes the aerodynamic damping coefficients, d_i represents the external disturbances. u_1 is the total lift force; u_ϕ , u_θ , u_ψ are the control inputs, which correspond to the roll, pitch, and yaw moments generated by the four propellers.

The model (1) contains six second-order subsystems with six degrees of freedom and four control inputs. It is a typical highly coupled nonlinear underactuated system. In order to achieve decoupling between position and attitude and reduce the complexity of controller design, the following vir-

tual control variables are defined as

$$u_x = \frac{u_1}{m}(\cos\phi\sin\theta\cos\psi + \sin\phi\sin\psi) \quad (2)$$

$$u_y = \frac{u_1}{m}(\cos\phi\sin\theta\sin\psi - \sin\phi\cos\psi) \quad (3)$$

$$u_z = \frac{u_1}{m}\cos\phi\cos\theta - g \quad (4)$$

To simplify the analysis, the dynamic model (1) can be rewritten as

$$\begin{cases} \dot{x}_{i1} = x_{i2} \\ \dot{x}_{i2} = u_i + f_i + d_i \\ y_i = x_{i1} \end{cases} \quad (5)$$

where $i = 1, 2, \dots, 6$; $x_{i1} = (x, y, z, \phi, \theta, \psi)$, $\dot{x}_{i1} = \dot{x}, \dot{y}, \dot{z}, \dot{\phi}, \dot{\theta}, \dot{\psi}$; y_i signifies the measurable output of (1). Let $y_{id} = (x_d, y_d, z_d, \phi_d, \theta_d, \psi_d)$ be the desired trajectory. The input signals of position and attitude can be expressed as $(u_1, u_2, u_3) = (u_x, u_y, u_z)$, $(u_4, u_5, u_6) = (\frac{1}{I_x}u_\phi, \frac{1}{I_y}u_\theta, \frac{1}{I_z}u_\psi)$; $(f_1, f_2, f_3) = (-\frac{k_x}{m}\dot{x}, -\frac{k_y}{m}\dot{y}, -\frac{k_z}{m}\dot{z})$, $(f_4, f_5, f_6) = (\frac{I_y - I_z}{I_x}\dot{\theta}\dot{\psi} - \frac{k_\phi}{I_x}\dot{\phi}, \frac{I_z - I_x}{I_y}\dot{\phi}\dot{\psi} - \frac{k_\theta}{I_y}\dot{\theta}, \frac{I_x - I_y}{I_z}\dot{\phi}\dot{\theta} - \frac{k_\psi}{I_z}\dot{\psi})$ are nonlinear terms in position and attitude, respectively; The external disturbances are indicated as $(d_1, d_2, d_3, d_4, d_5, d_6) = (d_x, d_y, d_z, d_\phi, d_\theta, d_\psi)$.

Control Objective: The control objective of this paper is to develop an adaptive prescribed-time controller to address the trajectory tracking problem of a QUAUV in the presence of external disturbances. The designed controller not only ensures the boundedness of all closed-loop signals but also guarantees that the QUAUV's position and attitude precisely track the desired trajectory y_{id} within the prescribed time.

Assumption 1 [21] For all $t > 0$, the desired trajectory y_{id} and its first-order derivative $\dot{y}_{id}$ are continuous and bounded.

2.2 Practical Prescribed-Time Stable

In existing time-varying control strategies, the gain functions are typically designed in a piecewise-continuous form, as shown in [20]. This form introduces time abrupt changes or non-smoothness issues. To overcome these problems, a novel time-varying gain function that combines exponential and polynomial functions is given by

$$\kappa(t) = \frac{1}{\varepsilon + (1 - \frac{t}{T})^n \cdot e^{-\lambda t}} \quad (6)$$

where T , ε , and λ are positive parameters, and n is a positive even integer. $0 \leq \varepsilon \leq 1$. Obviously, $\kappa(t)$ satisfies the following properties:

- 1) $\kappa(t)$ is a continuously increasing function on $[0, +\infty]$, and $\kappa(0) = 1/(1 + \varepsilon)$ and $\kappa(T) = 1/\varepsilon$.
- 2) $\kappa(t)$ has a minimum upper bound on $(T, +\infty)$, i.e., $\sup_{t > T} \kappa(t) = 1/\varepsilon$.
- 3) $\kappa(t)$ is C^n on $[0, +\infty]$.

Remark 1 In existing prescribed-time control methods, the time-varying gain function is usually in a piecewise continuous form [20], its specific expression is as follows:

$$\kappa(t) = \begin{cases} \frac{1}{\varepsilon + (1 - \frac{t}{T})^n}, & t \in [0, T] \\ \frac{1}{\varepsilon}, & t \in (T, +\infty) \end{cases} \quad (7)$$

The proposed non-piecewise time-varying gain function is specifically tailored to meet the stringent safety and stability requirements of power grid inspection. By ensuring continuity across the time domain, it resolves the chattering and discontinuity issues typical of traditional piecewise designs. This smooth analytical form effectively eliminates singularities and abrupt jumps, preventing mechanical jitter to guarantee actuator safety and the high-precision stability required for clear imagery.

Before achieving PPTS, some essential lemmas are given. Consider the nonlinear system as

$$\dot{x} = f(t, x) \quad (8)$$

where $x \in \mathbb{R}$ is the system state and $x(0) = x_0$, $f \in \mathbb{R}$ is a continuous function for x with $f(t, 0) = 0$. Set the state of the system (8) be $x(t; x_0)$.

Lemma 1 [22] For the system (8), if there exists a Lyapunov function $V(t)$ satisfying

$$\dot{V}(t) \leq -c\kappa(t)V(t) + \Delta \quad (9)$$

where c and Δ are positive constants, $\kappa(t)$ is a time-varying gain function, then the system (5) satisfies PPTS.

Lemma 2 [23] For any $x \in \mathbb{R}$ and $\epsilon > 0$, the following inequality holds

$$0 \leq |x| - x \tanh\left(\frac{x}{\epsilon}\right) \leq \delta\epsilon \quad (10)$$

where δ is a constant satisfying $\delta = e^{-\delta+1}$, i.e. $\delta = 0.2785$.

3 Main Results

In this section, a novel prescribed-time adaptive control strategy based on a time-varying gain function is proposed for QUAUV systems. Simultaneously, an adaptive law is designed to estimate the upper bound of the total disturbance online, thereby achieving reliable control of power grid inspection scenarios without relying on external disturbances or prior knowledge of model uncertainties.

3.1 Adaptive prescribed-time control design

Define the tracking error of QUAUV as

$$e_i = x_{i1} - y_{id} \quad (11)$$

and define a virtual error variable

$$z_i = x_{i2} - \alpha_i \quad (12)$$

where α_i is a virtual control input, which is defined as

$$\alpha_i = -c_1\kappa(t)e_i + \dot{y}_{id} \quad (13)$$

Then, the derivative of the tracking error (11) is given by

$$\begin{aligned} \dot{e}_i &= x_{i2} - \dot{y}_{id} \\ &= z_i + \alpha_i - \dot{y}_{id} \\ &= -c_1\kappa(t)e_i + z_i \end{aligned} \quad (14)$$

where c_1 is a positive parameter. The derivative of the virtual error variable (12) is given by

$$\begin{aligned} \dot{z}_i &= \dot{x}_{i2} - \dot{\alpha}_i \\ &= -c_2\kappa(t)z_i - \chi_i(t) - e_i - \hat{b}_m\gamma_i \tanh\left(\frac{\hat{b}_m z_i}{\epsilon}\right) \end{aligned} \quad (15)$$

where c_2 is a positive parameter, and $\chi_i(t) = f_i + d_i$ is the total disturbance resulting from the sum of external disturbances and model uncertainties.

Considering the given model assumptions and applying the absolute value inequality theorem, it is reasonable to assume the existence of positive unknown and bounded constants b_m, b_1, b_2 , and b_3 . As a result, the following inequality can be derived.

$$\begin{aligned} |\chi_i(t)| &\leq b_1 + b_2\dot{x}_{i1} + b_3\ddot{x}_{i1} \\ &\leq b_m\gamma_i \end{aligned} \quad (16)$$

where $\gamma_i = 1 + \dot{x}_{i1} + \ddot{x}_{i1}$, and $b_m = \max\{b_1, b_2, b_3\}$.

The practical prescribed-time controller u_i is designed as

$$u_i = -c_2\kappa(t)z_i - \chi_i(t) + \dot{\alpha}_i - e_i - \hat{b}_m\gamma_i \tanh\left(\frac{\hat{b}_m z_i}{\epsilon}\right) \quad (17)$$

where $\hat{b}_m$ is the estimation of b_m .

The adaptive updating law $\hat{b}_m$ is designed as

$$\dot{\hat{b}}_m = -c_3\kappa(t)\hat{b}_m + \gamma_i|z_i| \quad (18)$$

where $\hat{b}_m(0) \geq 0$, c_3 is a positive parameter.

Remark 2 The adaptive updating law in (18) is introduced to estimate the unknown upper bound of the lumped disturbance online, rather than requiring prior knowledge of the disturbance magnitude. Specifically, the term $\gamma_i|z_i|$ increases the estimate $\hat{b}_m$ when the virtual tracking error becomes large, thereby strengthening the disturbance-compensation capability of the controller. In contrast, the term $-c_3\kappa(t)\hat{b}_m$ acts as a forgetting mechanism, which suppresses excessive growth of the adaptive parameter and helps avoid overestimation when the tracking error gradually decreases.

In summary, a prescribed-time adaptive position and attitude tracking control framework is developed to the sum of external disturbances and model uncertainties to ensure prescribed performance. The overall control architecture is schematically illustrated in Fig. 2.

3.2 Stability analysis

Theorem 1 Consider the QUAUV dynamic system subject to the conditions outlined in Assumption 1. Through the synergistic implementation of the non-singular prescribed-time virtual control law (13), the prescribed-time trajectory tracking controller (17), and the adaptive updating law (18), the closed-loop system is rigorously proven to be PPTS. Specifically, this guarantees that all system signals remain uniformly bounded, and the tracking error is forced to converge to a small residual set containing the origin strictly within the user-defined prescribed time.

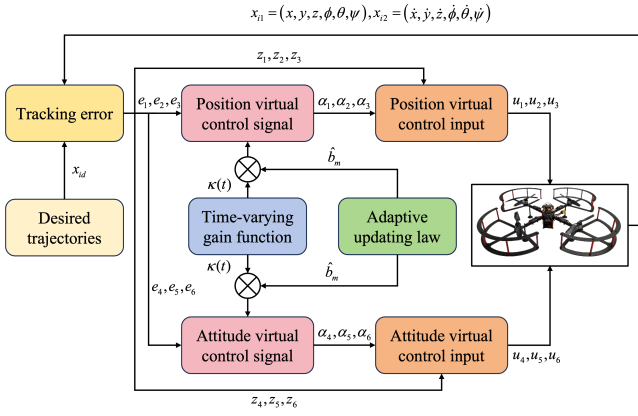


Fig. 2: Control block diagram.

Proof 1 The proof is given in detail as follows.

Step 1: Choose a positive Lyapunov function as

$$V_1 = \frac{1}{2}e_i^2 + \frac{1}{2}z_i^2 \quad (19)$$

Differentiating (19) results in

$$\dot{V}_1 = e_i \dot{e}_i + z_i \dot{z}_i \quad (20)$$

Taking (14) and (15) into (20) leads to

$$\begin{aligned} \dot{V}_1 = & e_i(-c_1\kappa(t)e_i + z_i) + z_i(-c_2\kappa(t)z_i - \chi_i(t) \\ & - e_i - \hat{b}_m\gamma_i \tanh(\frac{\hat{b}_m z_i}{\epsilon})) \end{aligned} \quad (21)$$

According to Lemma 2, one can obtain

$$\begin{aligned} \dot{V}_1 = & -c_1\kappa(t)e_i^2 - c_2\kappa(t)z_i^2 + |z_i||\chi_i(t)| \\ & - \hat{b}_m\gamma_i|z_i| + 0.2785\epsilon \end{aligned} \quad (22)$$

Step 2: Choose a positive Lyapunov function as

$$V_2 = V_1 + \frac{1}{2}\tilde{b}_m^2 \quad (23)$$

where $\tilde{b}_m = b_m - \hat{b}_m$.

Differentiating (23) results in

$$\dot{V}_2 = \dot{V}_1 - \tilde{b}_m \dot{\hat{b}}_m \quad (24)$$

Substituting (18) into (24) yields

$$\begin{aligned} \dot{V}_2 = & -c_1\kappa(t)e_i^2 - c_2\kappa(t)z_i^2 + |z_i||\chi_i(t)| \\ & - \hat{b}_m\gamma_i|z_i| + 0.2785\epsilon - \tilde{b}_m \dot{\hat{b}}_m \\ = & -c_1\kappa(t)e_i^2 - c_2\kappa(t)z_i^2 + |z_i||\chi_i(t)| \\ & - b_m\gamma_i|z_i| + \hat{b}_m(\gamma_i|z_i| - \dot{\hat{b}}_m) + 0.2785\epsilon \\ = & -c_1\kappa(t)e_i^2 - c_2\kappa(t)z_i^2 + c_3\kappa(t)\tilde{b}_m\hat{b}_m \\ & + 0.2785\epsilon \end{aligned} \quad (25)$$

By utilizing Young's inequality, one can obtain

$$\begin{aligned} \tilde{b}_m\hat{b}_m = & \tilde{b}_m b_m - \tilde{b}_m^2 \\ \leq & -\frac{1}{2}\tilde{b}_m^2 + \frac{1}{2}b_m^2 \end{aligned} \quad (26)$$

Substituting (26) into (25) yields

$$\begin{aligned} \dot{V}_2 = & -c_1\kappa(t)e_i^2 - c_2\kappa(t)z_i^2 - \frac{1}{2}c_3\kappa(t)\tilde{b}_m^2 \\ & + \frac{1}{2}c_3\kappa(t)b_m^2 + 0.2785\epsilon \\ \leq & -c_1\kappa(t)e_i^2 - c_2\kappa(t)z_i^2 - \frac{1}{2}c_3\kappa(t)\tilde{b}_m^2 \\ & + \frac{1}{2(1+\epsilon)}c_3b_m^2 + 0.2785\epsilon \\ = & -c\kappa(t)V_2 + \Delta \end{aligned} \quad (27)$$

where $c = \min\{c_1, c_2, \frac{1}{2}c_3\}$; $\Delta = \frac{1}{2(1+\epsilon)}c_3b_m^2 + 0.2785\epsilon$.

According to Lemma 1, if we choose sufficiently small ϵ and appropriate c, Δ , it can be obtained that the closed-loop system of the QUAV is PPTS, and converge to a small bounded set $V_2(t) \leq \max\{\frac{2\Delta\epsilon}{c}, \frac{\Delta}{c} + \epsilon_0\}$.

The proof of Theorem 1 is completed.

4 Simulation Results

This paper verifies the effectiveness of the proposed prescribed-time adaptive trajectory tracking control scheme through numerical simulation. The simulation was implemented using MATLAB R2022b/Simulink on an Intel Core computer equipped with a 2.5 GHz i5-12400F processor and running Windows 11.

The corresponding model parameters for the QUAV are given as: $m = 2\text{kg}$, $g = 9.8\text{m/s}^2$, $I_x = I_y = I_z = 1.745 \times 10^{-2}\text{kg} \cdot \text{m}^2$, $k_x = k_y = k_z = k_\phi = k_\theta = k_\psi = 0.008$, and the disturbances are assumed as $d_x = d_y = d_z = 0.1\sin(0.2t)$, $d_\phi = d_\theta = d_\psi = 0.2\sin(0.1t)$. The desired trajectories of the QUAV are selected as $x_d = 2\cos(t)$, $y_d = 2\sin(t)$, $z_d = 0.1t$, $\phi_d = \theta_d = \psi_d = 2\sin(t)$. For position subsystem, the initial values of the QUAV are chosen as $[x(0), y(0), z(0)] = [0.1, 0.2, -0.3]^T$. For attitude subsystem, the initial values are selected as $[\phi(0), \theta(0), \psi(0)] = [0.3, 0.4, -0.5]^T$. The initial values of adaptive updating law are set as $\hat{b}_m(0) = 0.5$. The time-varying gain function parameters are set as $\epsilon = 0.01$, $n = 2$, $\lambda = 0.01$, $T = 1$. The control parameters are set as $c_1 = c_2 = c_3 = 1$, $\epsilon = 0.01$.

Remark 3 The controller parameters are selected according to the following practical guidelines. The prescribed-time parameter T determines the desired convergence horizon and should be chosen based on the task requirement. The parameter ϵ affects the upper bound of the time-varying gain after T : a smaller ϵ generally strengthens the correction capability, but may also enlarge the control magnitude. Therefore, ϵ should be selected by balancing convergence speed and control smoothness. The even integer n and the coefficient λ jointly shape the evolution profile of the gain function $\kappa(t)$: n mainly affects the variation rate near the prescribed time, whereas λ is introduced to smooth the transient gain evolution and avoid overly abrupt changes. The gains c_1 and c_2 respectively determine the convergence intensity of the tracking error and the damping level of the virtual error, while c_3 adjusts the adaptation rate of the disturbance-bound estimate.

To verify the effectiveness of the proposed scheme, we conducted simulations. The corresponding simulation results are shown in Fig. 3 - Fig. 8. Fig. 3 shows that the actual

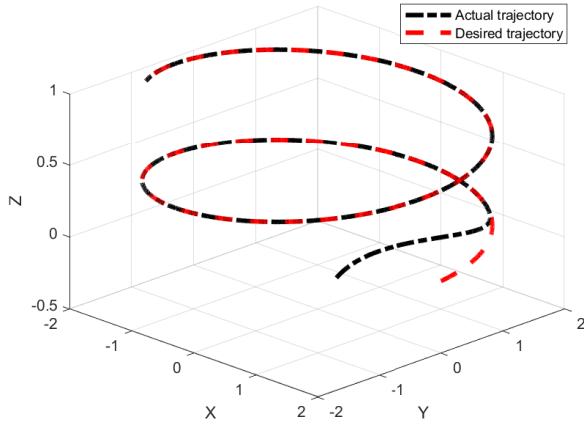


Fig. 3: The curves of trajectory tracking in 3D space.

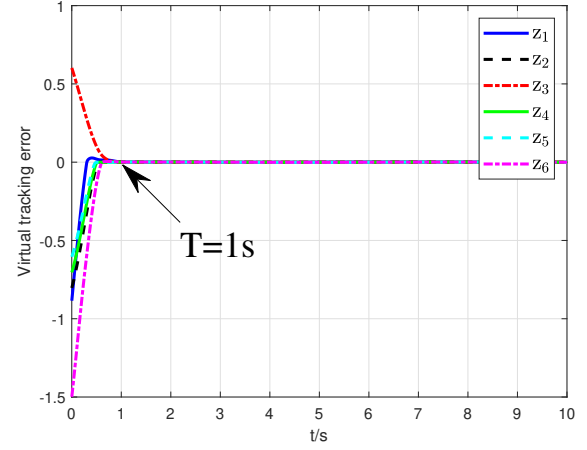


Fig. 6: The curves of the virtual tracking errors.

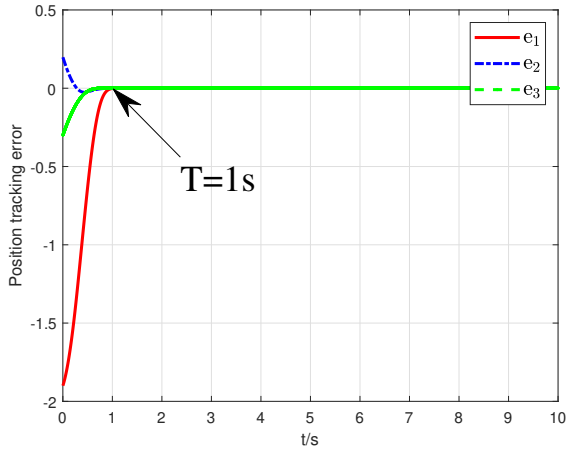


Fig. 4: The curves of position tracking errors.

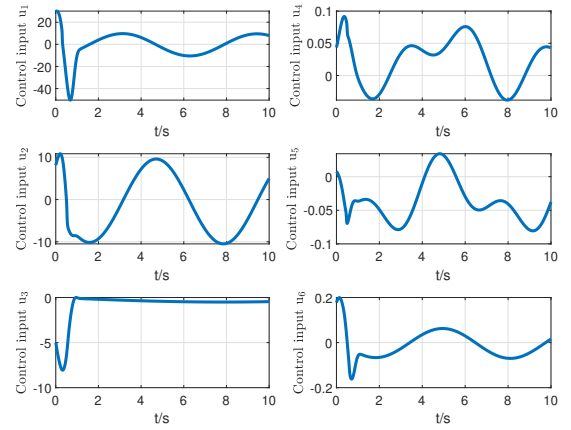


Fig. 7: The curves of attitude and position control input.

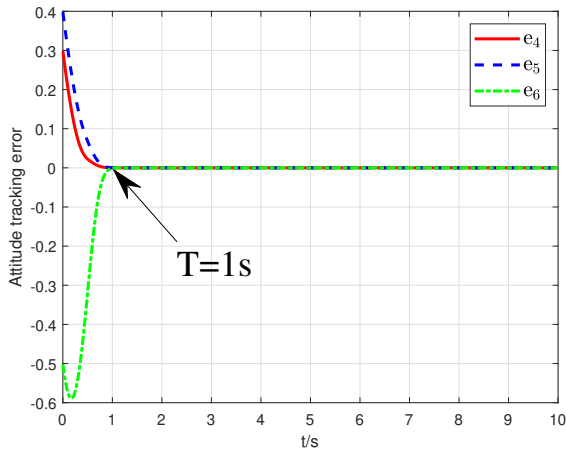


Fig. 5: The curves of attitude tracking errors.

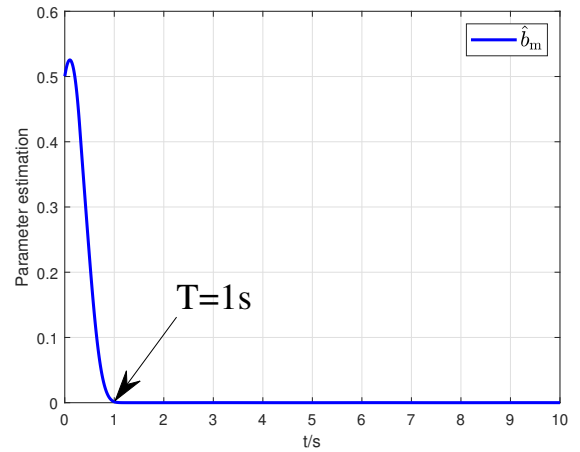


Fig. 8: The curves of the adaptive update law $\hat{b}_m$.

position and attitude trajectories accurately tracked the spiral 3D reference path of the desired trajectory within a specified time $T = 1s$, demonstrating excellent spatial tracking capability. The tracking errors in Fig. 4 and Fig. 5 further quantify this performance. Both position and attitude tracking errors converge rapidly to a steady state within 1s, and

$|e_i| \leq 1.7 \times 10^{-5}$, proving that the controller has a fast transient response within the specified time $T = 1s$. In addition, Fig. 6 shows the curves of the virtual tracking error. Under the action of the controller, the virtual tracking error decays rapidly within 1s, which also verifies that the proposed control strategy can guarantee the system has good conver-

gence performance and steady-state accuracy. Fig. 7 shows that the control input remains bounded and smooth throughout the operation, without obvious chattering, indicating that the proposed control law is physically feasible and beneficial for practical applications. Fig. 8 shows the adaptive law estimation curve, which effectively compensates for the inherent uncertainty of the model and external disturbances in the initial stage, demonstrating a rapid convergence process. After a prescribed time $T = 1s$, the curve stably converges to near the zero domain, highlighting the effectiveness of the adaptive mechanism.

5 Conclusion

This paper addresses the prescribed-time position and attitude trajectory tracking control problem of a QUAV in power system inspection scenarios. A non-piecewise time-varying gain function combining polynomial and exponential forms is constructed to solve the abrupt changes and non-smoothness introduced by piecewise operations. By combining an adaptive update law to estimate the uncertainties of the QUAV model and external disturbances, a prescribed-time controller is systematically developed, enabling the tracking error to converge to near the origin within a specified time, thereby improving the inspection efficiency of the QUAV and meeting the diverse inspection task requirements. Simulation results show that the proposed method has advantages such as fast convergence speed, high tracking accuracy, and user-specified convergence time.

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